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Dosing equipment

Abstract

A dosing equipment, including: a container; a dosing apparatus, at least partially provided within the container and connecting with an interior and an exterior of the container to convey the reagent from the interior of the container to the exterior of the container; a connector structure connecting the container opening and the dosing apparatus. The connector structure includes a connector penetrating along the axial direction of the container opening, a support member and a bearing member sequentially socketed within the connector. The connector is socketed on an outer periphery of the container opening, and a gap exists between the connector and the support member to receive a sidewall of the container opening. The bearing member is provided outside the dosing apparatus. A first sealing ring and a second sealing ring are provided to seal-connect the connector structure, the dosing apparatus, and the container opening, respectively.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to the technical field of weighing and sampling, and in particular relates to a dosing equipment.
BACKGROUND OF THE INVENTION
(2) It is well known that in biochemical experiments, weighing and sampling operations of reagent,

especially powder reagent, need to be performed frequently. However, the weighing balance commonly used at present not only requires manual operation, but also has low accuracy and large errors. Moreover, due to the manual operation, each operation is based on human handling to get a sample, so it may be difficult to achieve desired weighing and sampling result after several operations. In addition, due to manual operation, moving the sample may also cause the reagent to fall or scattered in the process, which will not only lead to waste of reagent, but may also cause reagent contamination. As a result, automated sampling devices are increasingly favored by experimentalists. However, existing automated sampling devices are complex in design and often very inconvenient to use.

SUMMARY OF THE INVENTION

(3) Embodiments of the present disclosure provide a dosing device, the dosing device has a structural design, by setting up a connector structure, not only the assembly between a container and a dosing apparatus can be realized, but also a sealed connection between the container and the dosing apparatus can be realized.

(4) Thus, the present disclosure provides the following embodiments:

(5) One embodiment of the present disclosure provides a dosing apparatus. The dosing apparatus includes: a container which is adapted to hold a reagent and includes a container opening; a dosing apparatus, which is at least partially provided within the container and is connecting with an interior and an exterior of the container to convey the reagent from the interior of the container to the exterior of the container; and a connector structure which is adapted to connect the container opening and the dosing apparatus. The connector structure includes a connector penetrating along the axial direction of the container opening, a support member and a bearing member sequentially socketed within the connector. The connector is socketed on an outer periphery of the container opening, and a gap is provided between the connector and the support member to receive a sidewall of the container opening, The bearing member is provided outside the dosing apparatus. A first sealing ring and a second sealing ring are provided between the support member and the bearing member and between the support member and the sidewall of the container opening, respectively, so as to seal-connect the connector structure, the dosing apparatus, and the container opening, respectively.

(6) Optionally, the dosing apparatus further includes a channel housing, and a restriction block disposed on an outer side of the channel housing; wherein an upper end surface of the bearing member is recessed downwardly with respect to an upper end surface of the support member, so as to be suitable for receiving and accommodating the restriction block, thereby defining a position at which the dosing apparatus is coupled to the container.

(7) Optionally, the dosing apparatus includes: a first channel, having a first channel first opening connecting with the container to allow the reagent inside the container to enter the first channel, and a first channel second opening which is in connection with outside of the first channel to allow the reagent to leave the first channel; and at least one first spiral rod provided within the first channel, wherein the least one first spiral rod is configured to be driven to rotate in a first direction, so as to deliver the reagent from the first channel to the first channel second opening, and wherein a mass flow rate of the reagent leaving through the first channel second opening is controlled by adjusting the rotational speed of the first spiral rod.

(8) Optionally, the dosing apparatus further includes a second channel, and at least one second spiral rod provided within the second channel; wherein the second channel has a second channel first opening in connection with the first channel second opening to allow the reagent within the first channel to enter the second channel, a second channel second opening in connection with the outside to allow the reagent to leave the second channel, and a second channel third opening in connection with the container to allow the reagent to leave the second channel and return to the container; wherein the at least one second spiral rod is in meshing transmission connection with the first spiral rod to rotate in a second direction as driven by the first spiral rod, to convey the reagent

within the second channel to the second channel second opening and the second channel third opening; and wherein the second direction is opposite to the first direction.
(9) Optionally, the first channel first opening is provided in a mid-side portion of the first channel; the first channel second opening is provided in a lower side portion of the first channel; the second channel first opening is provided in a lower side portion of the second channel; the second channel second opening is provided at a bottom end of the second channel; and the second channel third opening is provided at an upper side portion of the second channel.
(10) Optionally, a tip of the first spiral rod extends outside the first channel; a tip of the second spiral rod extends outside the second channel; the dosing apparatus comprises an end-face cam coaxially connected with the tip of the first spiral rod and a first gear coaxially connected with the tip of the second spiral rod; the end-face cam is in meshing transmission connection with the first gear; the first spiral rod is adapted to be driven by a driving apparatus to rotate in the first direction, thereby driving the second spiral rod to rotate in the second direction.
(11) Optionally, the dosing apparatus further includes a driving apparatus; the driving apparatus is connected to the first spiral rod and is adapted to drive the first spiral rod to rotate in the first direction, and to drive the second spiral rod through the first spiral rod to rotate in the second direction at the same time.
(12) Optionally, the driving apparatus further includes a drive housing; the drive housing having an opening matching the connector so as to accept and connect to the connector.
(13) Optionally, the dosing apparatus further includes a lifting and turning structure; the lifting and turning structure is coupled to the dosing apparatus to control a height and an angle of the dosing apparatus; and the dosing apparatus is adapted to be aligned with a reagent bottle for receiving the reagent when the dosing apparatus is outputting the reagent.
(14) Optionally, the lifting and turning structure is connected to the driving apparatus through the drive housing.
(15) The technical solution of the embodiments of the present disclosure has beneficial effects compared to the existing techniques.
(16) For example, the dosing apparatus has a simple structure, by providing the connector structure, not only an assembly between the container and the dosing apparatus can be easily constructed, but also a sealed connection between the container and the dosing apparatus can be realized.
(17) For another example, by providing the connector structure, not only can the connection position between the dosing apparatus and the container be well defined, but also the dosing apparatus and the container can be easily assembled, which is conducive to maintaining the stability of the assembly.
(18) Another example is that, by providing the connector structure, it becomes convenient for installation and disassembly between the dosing apparatus and the container, as well as between the dosing apparatus and the driving apparatus, speeding up replacement of the container, the dosing apparatus and any samples. This technique not only facilitates rapid changing reagent in need to weigh a variety of reagent, but also avoids cross-contamination and the burden of cleaning weighing apparatus caused by using the same weighing apparatus to weigh a variety of reagent, therefore saving time and labor.
(19) Another example is that, by providing the connector structure, the dosing apparatus and the drive housing can be connected, and by controlling the drive housing through the lifting and turning structure, the height and angle adjustment of the dosing apparatus can be realized. In this case, the dosing apparatus and the lifting and turning structure are not directly connected to each other, which will not affect the assembly between the dosing apparatus and the drive housing.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a cross-sectional view of one type of the dosing apparatus in an embodiment of the present disclosure;

(2) FIG. **2** is a partial schematic diagram of the dosing apparatus in an embodiment of the present disclosure;

(3) FIG. **3** is another partial schematic diagram of the dosing apparatus in an embodiment of the present disclosure;

(4) FIG. **4** is a partial cross-sectional view of the dosing apparatus in an embodiment of the present disclosure;

(5) FIG. **5** is a schematic view of one structure of the end-face cam in an embodiment of the present disclosure;

(6) FIG. **6** is a schematic view of one structure of the cam fitting member in an embodiment of the present disclosure;

(7) FIG. **7** is a schematic diagram of the outlet valve in an embodiment of the present disclosure, wherein the outlet valve is in a closed state;

(8) FIG. **8** is another schematic diagram of the outlet valve in an embodiment of the present disclosure, wherein the outlet valve is in an open state;

(9) FIG. **9** is a partial schematic diagram of the dosing head in an embodiment of the present disclosure, wherein the outlet valve is in a closed state;

(10) FIG. **10** is another partial schematic diagram of the dosing head in an embodiment of the present disclosure, wherein the outlet valve is in an open state;

(11) FIG. **11** is a third partial schematic view of the dosing apparatus in an embodiment of the present disclosure, wherein, for the third channel, only the lower portion thereof is shown and the upper portion thereof is not shown;

(12) FIG. **12** is a cross-sectional view of one embodiment of the dosing set in an embodiment of the present disclosure;

(13) FIG. **13** is a partial cross-sectional view of the dosing set in an embodiment of the present disclosure;

(14) FIG. **14** is a schematic diagram of the dosing head in an embodiment of the present disclosure.

(15) FIG. **15** is a third partial schematic view of the dosing head in an embodiment of the present disclosure.

(16) FIG. **16** is a schematic view of the locking structure in an embodiment of the present disclosure, wherein only a closed state of the locking structure is shown, and a state in which the locking structure locks the dosing apparatus is not shown;

(17) FIG. **17** is another schematic diagram of the locking structure in an embodiment of the present disclosure, wherein the locking structure is in an open state;

(18) FIG. **18** is a cross-sectional view of one embodiment of the dosing head in an embodiment of the present disclosure;

(19) FIG. **19** shows a schematic structural diagram of a dosing equipment in an embodiment of the present disclosure; and

(20) FIG. **20** is a schematic block diagram of a dosing system in an embodiment of the present disclosure.

REFERENCE NUMERALS

(21) **100** dosing apparatus, **111** first channel, **111***a* first channel first opening, **111***b* first channel second opening, **112** first spiral rod, **113** second channel, **113***a* second channel first opening, **113***b* second channel second opening, **113***c* second channel third opening, **114** second spiral rod, **115** end-face cam, **115***a* first beveled surface, A the high point of the first beveled surface, B the low point of the first beveled surface, **115***b* first vertical surface, **116** first gear, **117** top cover, **118** elastic member, **119** cam fitting member, **119***a* second beveled surface, C the high point of second

beveled surface, D the low point of second beveled surface, **119***b* second vertical surface, **119***c* guiding block, **119***d* support edge, **120** outlet valve, **121** closure portion, **122** hook portion, **122***a* hook slot, **123** hook block, **124** channel housing, **124***a* restriction block, **125** trigger section, **126** third channel, **126***a* third channel first opening, **127** conveyor belt, **127***a* first conveyor segment, **127***b* second conveyor segment, **127***c* groove, **200** dosing set, **211** container, **211***a* sidewall of container mouth, **220** connector structure, **221** connector, **222** support member, **223** bearing member, **224** first sealing ring, **225** second sealing ring, **300** dosing head, **310** driving apparatus, **311** drive motor, **312** second gear, **313** third gear, **314** fourth gear, **315** drive housing, **316** sensing structure, **317** outlet valve motor, **318** connecting rod, **320** locking structure, **321** toggle member, **322** cam, **322***a* camshaft, **323** abutment member, **323***a* extension portion, **323***b* opening end, **323***c* closing end, **324***a* first spring, **324***b* second spring, **400** dosing equipment, **410** lifting and turning structure, **500** dosing system, **510** balance, and **520** controller.

DETAILED DESCRIPTION OF THE INVENTION

(22) In order that the objects, features and beneficial effects of the present disclosure can be more obvious and understandable, specific embodiments of the present disclosure are described in detail below in conjunction with the accompanying drawings. It is to be understood that the specific embodiments described hereinafter are only for explaining the present disclosure, and are not intended to be a limitation of the present disclosure. Moreover, descriptions of the same and similar components in different embodiments and descriptions of components, features, effects, etc., which are part of the prior art may be omitted.

(23) Moreover, for the convenience of description, only part but not all of the structure related to the present disclosure may be shown in the accompanying drawings. Moreover, the same and similar attachment marks may be used in the accompanying drawings to refer to the same and similar parts in different embodiments.

(24) Referring to FIGS. **1** through **20**, embodiments of the present disclosure provide a dosing apparatus **100**, a dosing set **200**, a dosing head **300**, a dosing equipment **400**, and a dosing system **500**.

(25) A first aspect of embodiments of the present disclosure is that a dosing apparatus **100** is provided.

(26) Specifically, the dosing apparatus **100** includes a conveying channel **111**/**126** and a conveying mechanism. Wherein, the conveying channel **111**/**126** has a conveying channel first opening **111***a*/**126***a* connecting with a container **211** on the outside to allow a reagent within the container **211** to enter the conveying channel **111**/**126**, and a conveying channel second opening **111***b* connecting with the outside of the conveying channel **111**/**126** to allow the reagent to leave the conveying channel **111**/**126**. The conveying mechanism is provided within the conveying channel **111**/**126** and is adapted to be driven to move to convey the reagent within the conveying channel **111**/**126** to the conveying channel second opening **111***b*, as well as to control a mass flow rate of the reagent exiting the conveying channel through the conveying channel second opening **111***b* by adjusting a speed of its own movement.

(27) Referring to FIGS. **1** to **3**, in some embodiments, the conveying channel **111**/**126** may include a first channel **111**, the conveying channel first opening **111***a*/**126***a* may include a first channel first opening **111***a*, and the conveying channel second opening **111***b* may include a first channel second opening **111***b*. Accordingly, the conveying mechanism includes at least one first spiral rod **112** disposed within the first channel **111**.

(28) In particular embodiments, each of the at least one first spiral rod **112** is adapted to be driven to rotate in the first direction to convey the reagent within the first channel **111** to the first channel second opening **111***b*, as well as to control a mass flow rate of the reagent leaving through the first channel second opening **111***b* by adjusting a rotational speed of itself.

(29) It will be appreciated that by adjusting the rotational speed of the first spiral rod **112**, it is possible to control the mass of the reagent conveyed by the first spiral rod **112** per unit of time, and

thereby control the mass flow rate of the reagent leaving the first channel **111**. The mass flow rate of the reagent leaving the first channel **111** is the mass of the reagent leaving through the first channel second opening **111***b* in a unit of time. And, when the mass flow rate of the output reagent is determined, the mass of the output reagent is obtained based on the time that the reagent is output.
(30) With the above technical solution, on the one hand, the mass flow rate of the sampled reagent can be controlled by adjusting the rotational speed of the first spiral rod **112** so as to accurately weigh and sample the reagent, especially the powder reagent; on the other hand, the automation of the reagent sampling is realized so as to avoid the weighing error and reagent contamination caused by manual weighing and sampling.
(31) In some embodiments, the relationship between the mass flow rate of various reagent and the rotational speed of the first spiral rod **112** can be obtained experimentally. That is, the mass flow rate of the various reagent can be measured separately at different rotational speeds of the first spiral rod **112**, so as to obtain the relationship between the mass flow rate of the various reagent and the rotational speed of the first spiral rod **112**. The specific experimental process can be realized by any conventional technical means known in the art, and will not be repeated herein.
(32) In some embodiments, the dosing apparatus **100** may further include a second channel **113** and at least one second spiral rod **114** disposed within the second channel **113**.
(33) Specifically, the second channel **113** has a second channel first opening **113***a* in connection with the first channel second opening **111***b* to allow the reagent within the first channel **111** to enter the second channel **113**, a second channel second opening **113***b* in connection with the outside to allow the reagent to exit the second channel **113**, and a second channel third opening **113***c* in connection with the container **211** to allow the reagent to exit the second channel **113** and return to the container **211**.
(34) In particular embodiments, the second spiral rod **114** is in meshing transmission connection with the first spiral rod **112** to rotate in a second direction driven by the first spiral rod **112**, to convey the reagent within the second channel **113** to the second channel second opening **113***b* and the second channel third opening **113***c*. The second direction is opposite to the first direction.
(35) It is to be understood that by adjusting the rotational speed of the first spiral rod **112**, the mass of the reagent conveyed by the first spiral rod **112** per unit of time can be controlled, thereby controlling the mass flow rate at which the reagent leaves the first channel **111** and enters the second channel **113**, and thereby controlling the mass flow rate at which the reagent leaves the second channel **113**. The mass flow rate of the reagent leaving the second channel **113** is the mass of the reagent leaving through the second channel second opening **113***b* in a unit of time.
(36) With the above technical solution, the excess reagent in the first channel **111** can be made to enter the second channel **113** and be sent back to the container **211** through the second spiral rod **114** in the second channel **113**, thereby avoiding the accumulation and clumping of the reagent in the first channel **111**, as well as the possible influence on the rotation and rotational speed of the first spiral rod **112** that may be brought about by the accumulation and clumping of the reagent, and thus enabling the reagent to be smoothly output.
(37) In some embodiments, the first channel first opening **111***a* can be provided in the mid-side portion of the first channel **111** and connected to the container **211**; the first channel second opening **111***b* can be provided in the lower side portion of the first channel **111**; the second channel first opening **113***a* may be provided at a lower side portion of the second channel **113** and is in connection with the first channel second opening **111***b*; the second channel second opening **113***b* may be provided at a bottom end of the second channel **113** and is in connection with the exterior of the dosing apparatus **100**; and the second channel third opening **113***c* may be provided at an upper side portion of the second channel **113** and is in connection with the container **211**.
(38) In this way, excess reagent can be caused to return to the container **211** to avoid reagent backlog within the dosing apparatus **100**, thereby facilitating reagent flow, and thereby facilitating

reagent output through the dosing apparatus **100**.

(39) In a specific implementation, the pitch, diameter and rotational speed of the first spiral rod **112** and the second spiral rod **114** can be customized based on the specifics of the reagent. Wherein, the specifics of the reagent may include the type of reagent, the particle size of the reagent, and the delivery dose of the reagent.

(40) In some embodiments, the pitch difference, the diameter difference, and the rotational speed difference between the first spiral rod **112** and the second spiral rod **114** are all greater than zero. In this way, it can be beneficial for the reagent to be output outwardly through this dosing apparatus **100**.

(41) In some embodiments, this dosing apparatus **100** may include a first spiral rod **112** and a second spiral rod **114**. Wherein, this first spiral rod **112** and this second spiral rod **114** are in meshing transmission connection. Specific examples may be shown with reference to FIGS. **1** and **2**.

(42) In other embodiments, the dosing apparatus **100** may include at least two first spiral rods **112** and one second spiral rod **114**. Wherein the second spiral rod **114** is in meshing transmission connection with one of the at least two first spiral rods **112**. Specific examples may be shown with reference to FIG. **3**, in which the dosing apparatus **100** includes two first spiral rods **112** and a second spiral rod **114**.

(43) In yet further embodiments, the dosing apparatus **100** may include one first spiral rod **112** and at least two second spiral rods **114**. Wherein the first spiral rod **112** is in meshing transmission connection with one of the at least two second spiral rods **114**. And the at least two second spiral rods **114** are synchronously rotationally connected such that the second spiral rod **114** that is in meshing transmission connection with the first spiral rod **112** drives the other second spiral rods **114** to synchronously rotate.

(44) In yet further embodiments, the dosing apparatus **100** may include at least two first spiral rods **112** and at least two second spiral rods **114**. Wherein, one of the at least two second spiral rods **114** is in meshing transmission connection with to one of the at least two first spiral rods **112**, and the at least two second spiral rods **114** are synchronously rotationally connected to each other such that the second spiral rod **114** that is in meshing transmission connection with the first spiral rod **112** drives the other second spiral rods **114** to synchronously rotate.

(45) In specific implementations, the synchronously rotating individual second spiral rods **114** may be connected in any manner known in the art, without limitation herein. For example, the synchronized rotation of the individual second spiral rods **114** may be realized using planetary gears.

(46) In some embodiments, the dosing apparatus **100** may include at least two first spiral rods **112** that are synchronously rotationally connected to each other.

(47) In specific embodiments, the synchronously rotating individual first spiral rods **112** may also be connected in any manner known in the art, without limitation herein. For example, the synchronously rotating individual first spiral rods **112** may also be connected using planetary gears.

(48) As previously described, the second spiral rod **114** is in meshing transmission connection with the first spiral rod **112** and is adapted to rotate in a second direction driven by the first spiral rod **112**. Wherein the second direction is opposite to the rotation direction of the first spiral rod **112**, i.e., the first direction.

(49) Referring to FIGS. **1** to **3**, in some embodiments, the first spiral rod **112** and the second spiral rod **114** in the meshing transmission connection may be disposed parallel to each other, a tip of the first spiral rod **112** extends outside of the first channel **111**, and the tip of the second spiral rod **114** extends outside of the second channel **113**.

(50) Accordingly, the dosing apparatus **100** includes an end-face cam **115** rotationally connected coaxially with the tip of the first spiral rod **112**, and a first gear **116** rotationally connected coaxially with the tip of the second spiral rod **114**.

(51) In particular embodiments, the end-face cam **115** is in meshing transmission connection with the first gear **116**. The first spiral rod **112** is adapted to be driven by a driving apparatus **310** external to the dosing apparatus **100** via its bottom end to rotate in a first direction while driving the end-face cam **115** to rotate in a first direction, thereby driving the first gear **116** to rotate in a second direction, thereby driving the second spiral rod **114** to rotate in a second direction.
(52) In some embodiments, the first direction may be a counterclockwise direction or a clockwise direction. Accordingly, the second direction may be a clockwise direction or a counterclockwise direction.
(53) With the above technical solution, by setting the first spiral rod **112** and the second spiral rod **114** as parallel to each other and engaging in a drive connection, so that the first spiral rod **112** can drive the second spiral rod **114** to rotate in the second direction when it rotates in the first direction, it not only realizes the synchronization of the sample-giving (i.e., the output of the reagent) and sample-returning (i.e., the reagent being returned to the container **211**), but also eliminates an additional drive source for the second spiral rod **114**, saving space and cost of the product.
(54) It will be appreciated that “sample-giving” refers to outputting the reagent in the container **211** outside of the container **211** via the dosing apparatus **100**, and “sample-returning” refers to sending the reagent in the first channel **111** back into the container **211** via the second spiral rod **114**.
(55) In some embodiments, the dosing apparatus **100** provided by embodiments of the present disclosure may further include a rotation limiting structure to limit the first spiral rod **112** and the second spiral rod **114** from rotating in the opposite direction.
(56) Referring to FIGS. **1** to **4**, in some embodiments, the dosing apparatus **100** further includes a top cover **117** disposed above the first channel **111** and the second channel **113**, and an elastic member **118** and a cam fitting member **119** disposed sequentially between the top cover **117** and the end-face cam **115**. Wherein, the elastic member **118** and the cam fitting member **119** are adapted to move only in the direction of the axis of the first spiral rod **112**.
(57) Specifically, the cam fitting member **119** is adapted to reciprocate along the axial direction of the first spiral rod **112** driven by the end-face cam **115** when the end-face cam **115** is rotating in the first direction, and to limit the rotation of the end-face cam **115** in the second direction when the end-face cam **115** is to rotate in the second direction. The elastic member **118** is adapted to compress as well as release compression under the action of the cam fitting member **119** during rotation of the end-face cam **115** in the first direction.
(58) In some embodiments, the cam fitting member **119** further includes a guiding block **119***c* provided at a side portion thereof. Accordingly, the side portion of the top cover **117** has a guide slot extending in the direction of the axis of the first spiral rod **112**. The guide slot is adapted to receive the guiding block **119***c* and allow the guiding block **119***c* to reciprocate therein in the axial direction of the first spiral rod **112**.
(59) In some embodiments, the elastic member **118** may include a spring. Accordingly, the tip of the first spiral rod **112** passes through the end-face cam **115** and the cam fitting member **119** in turn and extends above the cam fitting member **119**. Moreover, the cam fitting member **119** has a support edge **119***d* provided on its inner ring to support the spring.
(60) In particular embodiments, the spring is sleeved on the outside of the first spiral rod **112** and its ends resist the inner end surface of the top cover **117** and the support edge **119***d*, respectively. In this way, the spring can be adapted to be compressed upwardly when the cam fitting member **119** is moved upwardly in the axial direction of the first spiral rod **112**, and to be released from compression downwardly when the cam fitting member **119** moves to the uppermost position, allowing the cam fitting member **119** to move downwardly in the axial direction of the first spiral rod **112**, thereby enabling the cam fitting member **119** to reciprocate in the axial direction of the first spiral rod **112**.
(61) Referring to FIGS. **5** and **6**, in some embodiments, the end-face cam **115** has a pair of first beveled surfaces **115***a* and a pair of first vertical surfaces **115***b* at one end thereof facing the cam

fitting member **119**. Correspondingly, the cam fitting member **119** has a pair of second beveled surfaces **119***a* and a pair of second vertical surfaces **119***b* at its end facing the end-face cam **115**. Wherein, the two first beveled surfaces **115***a* each covers a half turn of the end-face cam **115**; and the two second beveled surfaces **119***a* each covers a half turn of the cam fitting member **119**.
(62) In some embodiments, a high point A of one of the pair of first beveled surfaces **115***a* is adjacent to a low point B of the other first beveled surface **115***a* and they are connected by one of the first vertical surfaces **115***b*, while a low point B of one of the first beveled surfaces **115***a* is adjacent to a high point A of the other first beveled surface **115***a* and they are connected by the other first vertical surface **115***b*.
(63) Accordingly, a high point C of one of the pair of second beveled surfaces **119***a* is adjacent to a low point D of the other second beveled surface **119***a* and they are connected by one of the second vertical surfaces **119***b*, while a low point D of one second beveled surface **119***a* is adjacent to a high point C of the other second beveled surface **119***a* and they are connected by the other second vertical surface **119***b*.
(64) In the initial state, the first beveled surfaces **115***a* are opposite to and mated with the second beveled surfaces **119***a*, and the first vertical surfaces **115***b* are opposite to and mated with the second vertical surfaces **119***b*. Moreover, when the first beveled surfaces **115***a* are mated with the second beveled surfaces **119***a*, the high point A of each first beveled surface **115***a* meets the low point D of the corresponding second beveled surface **119***a*, and the low point B of each first beveled surface **115***a* meets the high point C of the corresponding second beveled surface **119***a*.
(65) When the end-face cam **115** is rotated in the first direction, each first vertical surface **115***b* will move away from the corresponding second vertical surface **119***b* with which it is mated, and the first beveled surface **115***a* may rotate relative to the corresponding second beveled surface **119***a* with which it is mated, and push the cam fitting member **119** to reciprocate along the direction of the axis of the first spiral rod **112**. At the same time, the elastic member **118** compresses as well as releases compression under the action of the cam fitting member **119**.
(66) In a specific implementation, when the end-face cam **115** rotates half a turn in the first direction, each first vertical surface **115***b* rotates in the direction of departing from the corresponding second vertical surface **119***b*, each first vertical surface **115***b* separates from the corresponding second vertical surface **119***b*, each first beveled surface **115***a* rotates with respect to the corresponding second beveled surface **119***a*, and a change from the high point A of each first beveled surface **115***a* meeting the low point D of the corresponding second beveled surface **119***a* to the high point A of each first beveled surface **115***a* meeting the high point C of the corresponding second beveled surface **119***a* occurs, thereby pushing the cam fitting member **119** upwardly in the direction of the axis of the first spiral rod **112**, while the elastic member **118** is compressed upwardly by the cam fitting member **119**.
(67) In particular embodiments, when the high point A of each first beveled surface **115***a* meeting the low point D of the corresponding second beveled surface **119***a* changes to the high point A of each first beveled surface **115***a* meeting the high point C of the corresponding second beveled surface **119***a*, the cam fitting member **119** is moved from the lowermost to the uppermost.
(68) When the end-face cam **115** continues to rotate half a turn in the first direction, each second beveled surface **119***a* disengages from the corresponding first beveled surface **115***a*, the upward force applied to the elastic member **118** is withdrawn, the elastic member **118** releases its compression downwardly and pushes the cam fitting member **119** downwardly in the direction of the axis of the first spiral rod **112** until each second beveled surface **119***a* once again mates with the corresponding first beveled surface **115***a*.
(69) With one rotation of the end-face cam **115** in the first direction, the cam fitting member **119** completes an up and down reciprocal movement in the axial direction of the first spiral rod **112**, and the elastic member **118** completes a compression and a release of that compression. When the end-face cam **115** continues to rotate in the first direction for more than one revolution, the cam

fitting member **119** completes more than one reciprocal movement along the axial direction of the first spiral rod **119**, and the elastic member **118** completes more than one compression and release of that compression.
(70) Since, in the initial state, each first beveled surface **115***a* is mated with the corresponding second beveled surfaces **119***a*, the first vertical surfaces **115***b* each is opposite to and mated with the corresponding second vertical surface **119***b*, and each second vertical surface **119***b* can only move upward and downward along the axial direction of the first spiral rod **112** and cannot rotate in the first direction or in the second direction, in the case that the end-face cam **115** is to rotate in the second direction, each first vertical surface **115** is mated with the corresponding second vertical surface **119***b*, and each second vertical surface **119***b* blocks the corresponding first vertical surface **115***b* from rotating in the second direction, thereby restricting the end-face cam **115** from rotating in the second direction.
(71) With the above technical solution, the first spiral rod **112** and the second spiral rod **114** are made suitable to rotate only in the direction along which the reagent is output, and the first spiral rod **112** and the second spiral rod **114** are restricted from rotating in the opposite direction. When the first spiral rod **112** and the second spiral rod **114** are forced to rotate in the opposite direction, the first spiral rod **112** and/or the second spiral rod **114** will be damaged due to the restriction of the cam fitting member **119**, which in turn will cause the dosing apparatus **100** to be destroyed. In this way, it is possible to make the dosing apparatus **100** not reusable, thereby making a dosing apparatus **100** usable only as a one-time dedicated delivery device for one kind of reagent, thereby avoiding the dosing apparatus **100** from being reused, and thereby preventing problems such as cross contamination of the reagent or reagent misuse due to reuse.
(72) It is understood that when the dosing apparatus **100** does not include a rotation limiting structure, the dosing apparatus **100** is reusable.
(73) Referring to FIGS. **2**, **3**, and **7** to **10**, the dosing apparatus **100** further includes an outlet valve **120** adapted to close as well as open the second channel second opening **113***b*, to effectively prevent leakage of the reagent in the dosing apparatus **100**.
(74) Specifically, the outlet valve **120** is rotationally connected to the end of the second channel **113** proximate to the second channel second opening **113***b*, and is adapted to be driven to rotate in a third direction to close the second channel second opening **113***b*, and to rotate in a fourth direction to open the second channel second opening **113***b*. Wherein, the fourth direction is opposite to the third direction.
(75) In some embodiments, the third direction may be a counterclockwise direction or a clockwise direction. Accordingly, the fourth direction may be a clockwise direction or a counterclockwise direction.
(76) In some embodiments, the outlet valve **120** includes a closure portion **121**. The closure portion **121** includes a sealing member adapted to be disposed facing the second channel second opening **113***b*. The sealing member is adapted to move closer to and face the second channel second opening **113***b* as the outlet valve **120** rotates in the third direction, so as to be suitable for closing the second channel second opening **113***b* by embedding in the second channel second opening **113***b*. The sealing member is also adapted to open the second channel second opening **113***b* by disengaging from and moving away from the second channel second opening **113***b* as the outlet valve **120** is rotated in the fourth direction.
(77) In some embodiments, the sealing member may be a silicone cap that fits over the second channel second opening **113***b* and is adapted to be embedded in the second channel second opening **113***b*, to close the second channel second opening **113***b*.
(78) In some embodiments, the dosing apparatus **100** further includes a hook block **123** disposed on an outer side of the second channel **113**. Accordingly, the outlet valve **120** further includes a hook portion **122** coupled to the closure portion **121** and bent relative to the closure portion **121** to be adapted to be disposed facing a side portion of the second channel **113**.

(79) Specifically, the hook portion **122** includes a hook slot **122***a* facing the hook block **123**. The hook slot **122***a* is adapted to move toward the hook block **123** when the outlet valve **120** is rotated in a third direction, and to be limited by the hook block **123** when moving to the hook block **123** so that the sealing member is set facing the second channel second opening **113***b* so as to be embedded in the second channel second opening **113***b*, as well as to move away from the hook block **123** when the outlet valve **120** is rotated in a fourth direction.
(80) In some embodiments, when the hook slot **122***a* is limited by the hook block **123**, the hook slot **122***a* abuts against the bottom of the hook block **123**. In this way, the upward movement of the outlet valve **120** can be restricted so that the sealing member is stably embedded within the second channel second opening **113***b*.
(81) Referring to FIGS. **9** and **10**, in some embodiments, the dosing apparatus **100** further includes a channel housing **124** adapted to at least partially harbor the first channel **111** and the second channel **113**. The first channel **111** and the second channel **113** are at least partially nested within the channel housing **124**. Further, the channel housing **124** has a trigger section **125** at a lower side of the channel housing **124** adapted to face a sensing structure **316** outside the channel housing **124**. The sensing structure **316** is coupled to the outlet valve **120** and is adapted to trigger movement of the outlet valve **120** in a third direction to close the second channel second opening **113***b* when it is in contact with the trigger section **125**, and to trigger movement of the outlet valve **120** in a fourth direction to open the second channel second opening **113***b* when it is out of contact with the trigger section **125**.
(82) In some embodiments, the dosing apparatus **100** further includes an outlet valve motor **317** coupled to the outlet valve **120** and the sensing structure **316**, respectively. The sensing structure **316** triggers the outlet valve motor **317** to control the movement of the outlet valve **120** in the third direction when the sensing structure **316** is in contact with the trigger section **125**, and the sensing structure **316** triggers the outlet valve motor **317** to control the movement of the outlet valve **120** in the fourth direction when the sensing structure **316** is out of contact with the trigger section **125**.
(83) In some embodiments, the sensing structure **316** may employ a microswitch.
(84) In some embodiments, the outlet valve motor **317** may employ a servo.
(85) In some embodiments, when the dosing apparatus **100** includes both a top cover **117** and a channel housing **124**, the top cover **117** may be mounted on top of the channel housing **124**.
(86) Referring to FIG. **11**, in other embodiments, the conveying channel **111**/**126** includes a third channel **126**, the conveying channel first opening **111***a*/**126***a* includes a third channel first opening **126***a*, and the conveying channel second opening **111***b* includes a third channel second opening.
(87) Accordingly, the conveying mechanism includes a conveyor belt **127** disposed within the third channel **126**; the conveyor belt **127** is adapted to be driven to drive, and to receive a reagent from the third channel first opening **126***a* during the driving process and to convey the reagent to the third channel second opening to output the reagent through the third channel second opening.
(88) In particular embodiments, the conveyor belt **127** includes a continuously varying first conveyor segment **127***a* and a second conveyor segment **127***b*. Wherein, the first conveyor segment **127***a* is adapted to convey the reagent to the third channel second opening; and the second conveyor segment **127***b* is adapted to return the reagent that does not exit through the third channel second opening back to the third channel **126**.
(89) It will be appreciated that the conveyor belt **127** is annular in shape and that any segment of the conveyor belt **127** is continuously changing in displacement during transmission, so that the position of the first conveyor segment **127***a* adapted to convey the reagent to the third channel second opening is continuously changing in the conveyor belt **127**, and the position of the second conveyor segment **127***b* adapted to return the reagent that has not left through the third channel second opening is continuously changing in the conveyor belt **127**.
(90) In some embodiments, the third channel **126** includes a third channel third opening in connection with the container **211** to allow the reagent that has not left through the third channel

second opening to return to the container **211**.
(91) In particular embodiments, the third channel first opening **126***a* may be disposed in a mid-side portion of the third channel **126** and in connection with the container **211** to receive the reagent within the container **211**; the third channel second opening may be disposed in a bottom portion of the third channel and in connection with the exterior of the dosing apparatus **100** to output the reagent; and the third channel third opening may be disposed in an upper side portion of the third channel and in connection with the container **211** so that the reagent in the third channel **126** can be returned to the container **211**.
(92) In this way, it is facilitated that the reagent within the container **211** enters the third channel **126** and is transported to the outside of the container **211** and back into the container **211** again via the conveyor belt **127** disposed within the third channel **126**, thereby facilitating the distribution and transport of the reagent through the dosing apparatus **100**.
(93) In some embodiments, a number of grooves **127***c* may also be provided on the conveyor belt **127** to facilitate reception and delivery of the reagent by the conveyor belt **127**.
(94) It should be noted that in the embodiments of the present disclosure, the volume of the conveying channel and the size of its individual openings can be customized, and the speed of movement of the conveyor mechanism within the conveying channel can be adjusted, so that the dosing apparatus **100** is suitable for sampling large or small doses of the reagent.
(95) In addition, the dosing apparatus in the current state of the art requires manual operation in small quantities for sampling doses of less than 10 mg, and the sampling results have low precision and large errors.
(96) In contrast, by using the dosing apparatus **100** provided in the embodiment of the present disclosure, accurate sampling for the reagent with a dose of 2 milligrams and below has been tested and proven.
(97) A second aspect of the embodiment of the present disclosure is to provide a dosing set **200**.
(98) Referring to FIG. **12**, the dosing set **200** includes a container **211** and a dosing apparatus **100**. Wherein the container **211** is adapted to hold the reagent. The dosing apparatus **100** is at least partially disposed within the container **211** and is in connection with an exterior of the container **211** to deliver the reagent within the container **211** to the exterior of the container **211**.
(99) In particular embodiments, the dosing apparatus **100** may include the dosing apparatus **100** provided in the first aspect of an embodiment of the present disclosure.
(100) Referring to FIGS. **12** and **13**, in some embodiments, the container **211** and the dosing apparatus **100** may be connected to each other by a connector structure **220**.
(101) Specifically, the container **211** includes a container opening. The connector structure **220** includes a connector **221** penetrating in an axial direction along container opening, and a support member **222** and a bearing member **223** that are sequentially set within the connector **221**.
(102) In particular embodiments, the dosing apparatus **100** is inserted in the bearing member **223**; the connector **221** is removably socketed around an outer periphery of the container opening; and there is a gap between the connector **221** and the support member **222** to receive the sidewall **211***a* of the container opening.
(103) Further, a first sealing ring **224** and a second sealing ring **225** are also provided between the support member **222** and the bearing member **223**, and between the support member **222** and the sidewall **211***a* of the container opening, respectively, to seal-connect the connector structure **220** to the dosing apparatus **100** and the container opening, respectively.
(104) In some embodiments, a threaded connection may be provided between the connector **221** and the outer periphery of the container opening.
(105) In some embodiments, the dosing apparatus **100** further includes a channel housing **124** adapted to at least partially harbor the first channel **111** and the second channel **113**, and the channel housing **124** is further provided with a restriction block **124***a* on an outer side of the channel housing **124** to define a position at which the dosing apparatus **100** is threaded in the

connector structure **220**, and thus a position at which the dosing apparatus **100** is coupled to the container **211**.
(106) In some embodiments, the upper end surface of the bearing member **223** may be recessed downwardly with respect to the upper end surface of the support member **222** and is adapted to receive as well as accommodate the restriction block **124***a*. At the same time, the first sealing ring **224** may be provided between the restriction block **124***a* and the upper end surface of the bearing member **223**.
(107) In this way, not only can the position of the dosing apparatus **100** threaded in the connector structure **220** can be defined to define the connection position between the dosing apparatus **100** and the container **211**, but also the dosing apparatus **100** can be stably fixed to the connector structure **220**, thereby providing a stable connection between the dosing apparatus **100** and the container **211**.
(108) In specific embodiments, the restriction block **124***a* may be provided in the middle of the channel housing **124** and located below the first channel first opening **111***a* so as not to interfere with the entry of the reagent in the container **211** into the first channel **111** through the first channel first opening **111***a*.
(109) In some embodiments, the first spiral rod **112** is adapted to be driven to rotate in a first direction at a first speed. At the same time, the container **211** is also adapted to be driven to rotate in the first direction at a second speed. Wherein, the second speed is less than the first speed.
(110) In this way, the reagent in the container **211** can be made to flow to prevent accumulation, thereby facilitating the reagent in the container **211** to smoothly enter the first channel **111** through the first channel first opening **111***a*, and thereby facilitating the reagent to be smoothly output through the dosing apparatus **100**.
(111) A third aspect of the embodiment of the present disclosure is to provide a dosing head **300**.
(112) Referring to FIGS. **14** to **18**, the dosing head **300** includes a dosing set **200** and a driving apparatus **310**. Wherein the dosing set **200** includes a container **211** and a dosing apparatus **100**; the container **211** is adapted to hold a reagent; the dosing apparatus **100** includes a first dosing mechanism and a second dosing mechanism that are in meshing transmission connection, wherein the first dosing mechanism is adapted to deliver the reagent within the container **211** to the second dosing mechanism, and the second dosing mechanism is adapted to deliver the reagent to the exterior of the container **211**; and the driving apparatus **310** is connected to the first dosing mechanism and is adapted to drive the first dosing mechanism to rotate in a first direction while driving the second dosing mechanism to rotate in a second direction through the first dosing mechanism; the second direction being opposite to the first direction.
(113) In particular embodiments, the first dosing apparatus includes at least a first channel **111** and a first spiral rod **112** provided in the first aspect of the embodiments of the present disclosure, and the second dosing apparatus includes at least a second channel **113** and a second spiral rod **114** provided in the first aspect of the embodiments of the present disclosure.
(114) In some embodiments, the drive mechanism **310** may include a drive motor **311**, a second gear **312** coaxially coupled to the drive motor **311**, and a third gear **313** in meshing transmission connection with the second gear **312**.
(115) In particular embodiments, the first spiral rod **112** is coaxially connected to the third gear **313**. The second gear **312** is adapted to rotate under the drive of the drive motor **311** and drive the third gear **313** to drive the first spiral rod **112** to rotate in the first direction at the first speed.
(116) It is to be understood that the coaxial connection of the second gear **312** with the drive motor **311** indicates that the second gear **312** is synchronously rotationally connected with the drive motor **311**; and the coaxial connection of the first spiral rod **112** with the third gear **313** indicates that the first spiral rod **112** is synchronously rotationally connected with the third gear **313**.
(117) As previously mentioned, in some embodiments, at least two first spiral rods **112** may be provided. In this case, the at least two first spiral rods **112** may be synchronously rotationally

connected, e.g., planetary gears may be used for synchronous rotational connection, and one of the at least two first spiral rods **112** is coaxially connected to the third gear **313** to drive that first spiral rod **112** to rotate in the first direction by the drive motor **311**, so as to drive the other first spiral rods by that first spiral rod **112** to rotate in the first direction.

(118) Further, the first spiral rod **112** coaxially connected to the third gear **313** may also be in meshing transmission connection with the second spiral rod **114**.

(119) As previously mentioned, in some embodiments, at least two second spiral rods **114** may be provided. In this case, the at least two second spiral rods **114** may be synchronously rotationally connected to each other, for example, planetary gears may be used for synchronous rotational connection, and one of the at least two second spiral rods **114** is in meshing transmission connection with a first spiral rod **112** coaxially connecting the third gear **313** to drive the second spiral rod **114** to rotate in the second direction by this first spiral rod **112**, thereby driving the other second spiral rod **114** to rotate in the second direction.

(120) As previously mentioned, in some embodiments, the first spiral rod **112** is adapted to be driven to rotate in the first direction at a first speed. The container **211** is also adapted to be driven to rotate in the first direction at the second speed. Wherein, the second speed is less than the first speed.

(121) In this case, the drive mechanism **310** may also include a fourth gear **314** that is in meshing transmission connection with the second gear **312**. And, the fourth gear **314** is coaxially connected to the container **211**. Wherein, a diameter ratio of the third gear **313** to the fourth gear **314** is equal to a speed ratio of the second speed to the first speed.

(122) In a specific implementation, the second gear **312** is adapted to rotate under the drive of the drive motor **311** and drive the third gear **313** to drive the first spiral rod **112** to rotate in the first direction at the first speed, and drive the fourth gear **314** to drive the container **211** to rotate in the first direction at the second speed.

(123) With the above technical solution, the container **211** and the dosing apparatus **100** can be driven to rotate at the same time by the same drive mechanism **310**, which not only saves the drive source, but also saves the occupying space and cost of the product.

(124) In some embodiments, the drive mechanism **310** may also include a drive housing **315** adapted to house the drive motor **311**, the second gear **312**, the third gear **313**, and the fourth gear **314**. And, the drive housing **315** may have an opening that mates with the connector structure **220** to connect to the connector structure **220**, and to connect the container **211**, the dosing apparatus **100**, and the drive mechanism **310** via the connector structure **220**.

(125) In some embodiments, the opening of the drive housing **315** may have dimensions that mate with an outer periphery of the connector **211** to receive the connector **211**, thereby connecting the container **211**, the dosing apparatus **100**, and the drive mechanism **310** via the connector structure **220**.

(126) In some embodiments, when the container **211** is simultaneously driven to rotate via the drive mechanism **310**, a rotational connection is employed between the opening of the drive housing **315** and the outer periphery of the connector **211**, for example, a bearing rotational connection may be employed.

(127) In other embodiments, when the container **211** is not rotating, a fixed connection between the opening of the drive housing **315** and the outer periphery of the connector **211** may be employed, for example, a threaded connection may be employed.

(128) In this embodiment of the present disclosure, the container **211** and the dosing apparatus **100**, as well as the dosing set **200** (including the container **211** and the dosing apparatus **100**) and the drive mechanism **310** can be conveniently mounted and dismounted via the connector structure **220**. It not only facilitates the rapid replacement of the container **211**, the dosing apparatus **100** and the dosing set **200** so that the reagent can be quickly replaced in the event that multiple reagent is to be weighed, saving time and labor, but also avoids the reagent contamination that may be brought

about when multiple reagent is weighed using the same weighing apparatus for sampling, as well as the trouble of cleaning the weighing apparatus.
(129) It can be understood that with the technical solution provided by the embodiment of the present disclosure, different kinds of reagent can be weighed and sampled using different containers **211** and the dosing apparatus **100** respectively, or using the same container **211** and the dosing apparatus **100** can be used for weighing and sampling. However, when the same container **211** and the same dosing apparatus **100** are used for weighing and sampling different reagent, different reagent needs to be replaced in the container **211**, and the residual reagent in the container **211** and the dosing apparatus **100** needs to be cleaned in a timely manner prior to each replacement of the reagent.
(130) In particular embodiments, a lower portion of the dosing apparatus **100** is adapted to be inserted into the drive housing **315** to coaxially connect the first spiral rod **112** to the third gear **313**, and to connect the container **211** to the fourth gear **314**.
(131) In some embodiments, the bottom end of the first spiral rod **112** has a plug. Accordingly, the third gear **313** has a socket that mates with the plug. The plug is tightly embedded into the socket to connect the first spiral rod **112** and the third gear **313**.
(132) In some embodiments, the drive mechanism **310** further includes a connecting rod **318** at least partially stowed within the drive housing **315**; one end of the connecting rod **318** is coaxially coupled to the fourth gear **314**, and the other end thereof is adapted to be coaxially coupled to the container **211** when the lower portion of the dosing apparatus **100** is inserted within the drive housing **315**.
(133) In some embodiments, the container **211** also has a container bore. The other end of the connecting rod **318** is adapted to be inserted and snapped into this container bore to connect the container **211** so as to drive the container **211** to rotate.
(134) In some embodiments, the drive housing **315** further has a through-hole suitable for the connecting rod **318** to pass through. The other end of the connecting rod **318** passes through the through-hole to connect to the container bore.
(135) In some embodiments, the dosing head **300** further includes a sensing structure **316** disposed within the drive housing **315**. The dosing apparatus **100** further includes a channel housing **124** adapted to at least partially harbor the first channel **111** and the second channel **113**. The first channel **111** and the second channel **113** are at least partially nested within the channel housing **124**. Moreover, the lower side portion of the channel housing **124** has a trigger section **125** adapted to be disposed facing the sensing structure **316**.
(136) In particular embodiments, the trigger section **125** contacts the sensing structure **316** when the lower portion of the dosing apparatus **100** is inserted within the drive housing **315**. The sensing structure **316** is coupled to the outlet valve **120** and is adapted to trigger movement of the outlet valve **120** in a third direction to close the second channel second opening **113***b* when the sensing structure **316** is in contact with the trigger section **125**, and to trigger movement of the outlet valve **120** in a fourth direction to open the second channel second opening **113***b* when the sensing structure **316** is out of contact with the trigger section **125**.
(137) In some embodiments, the dosing head **300** further includes an outlet valve motor **317** coupled to the outlet valve **120** and the sensing structure **316**, respectively. When the sensing mechanism **316** is in contact with the trigger section **125**, the outlet valve motor **317** is triggered to control the movement of the outlet valve **120** in the third direction; when the sensing structure **316** is out of contact with the trigger section **125**, the outlet valve motor **317** is triggered to control the movement of the outlet valve **120** in the fourth direction.
(138) Referring to FIGS. **16** and **17**, the dosing head **300** further includes a locking structure **320** to lock the dosing apparatus **100** when the dosing apparatus **100** is inserted into the drive housing **315**.
(139) In some embodiments, the locking structure **320** may include a toggle member **321**, a cam **322**, an abutment member **323**, a first spring **324***a*, and a second spring **324***b*. Among other things,

the toggle member **321** is disposed on an outer side of the drive housing **315** and is coupled to the cam **322** through an inner sidewall thereof. The cam **322** is rotatably coupled to the inner side of the drive housing **315** by the camshaft **322***a*; and the portion of the cam **322** remote from the toggle member **321** is adapted to abut against the outer side of the abutment member **323** and to move along the outer side of the abutment member **323**. The inner side of the abutment member **323** has an extension portion **323***a*; a first spring **324***a* and a second spring **324***b* are provided on both sides of the extension portion **323***a*, respectively.
(140) Accordingly, the locking structure **320** further includes a receiving slot provided on the outer side of the dosing apparatus **100** and receiving holes disposed on both sides of the receiving slot, which may be, for example, a receiving slot on the outer side of the channel housing **124** and receiving holes disposed on both sides of the receiving slot.
(141) In some embodiments, the receiving slots and receiving holes may be recessed inwardly with respect to the outer side of the dosing apparatus **100** (e.g., the outer side of the channel housing **124**) to flatten the outer side of the dosing apparatus **100** (e.g., the outer side of the channel housing **124**) for aesthetics.
(142) In particular embodiments, the receiving slot is adapted to receive and accommodate the extension portion **323***a* of the abutment member **323**. Two receiving holes on both sides of the receiving slot are adapted to receive a first spring **324***a* and a second spring **324***b*, respectively. Both ends of the first spring **324***a* are coupled to the inside of one end of the abutment member **323** and the inside of a first receiving hole, respectively, and both ends of the second spring **324***b* are coupled to the inside of the other end of the abutment member **323** and the inside of a second receiving hole, respectively.
(143) In some embodiments, the two ends of the abutment member **323** may be referred to as the opening end **323***b* and the closing end **323***c*, respectively, and the width of the abutment member **323** gradually increases along the direction of the opening end **323***b* pointing toward the closing end **323***c*.
(144) In particular embodiments, the first spring **324***a* is attached to the inner side of the opening end **323***b*, and the second spring **324***b* is attached to the inner side of the closing end **323***c*.
(145) When the dosing apparatus **100** is inserted into the drive housing **315**, the cam **322** may abut against the outer side of the opening end **323***b* of the abutment member **323**. When the toggle member **321** is pivoted towards the direction of the closing end **323***c* of the abutment member **323**, the cam **322** rotates about the camshaft **322***a* and moves from the outer side of the opening end **323***b* of the abutment member **323** to the outer side of the closing end **323***c* of the abutment member **323**.
(146) Since the width of the closing end **323***c* of the abutment member **323** is thicker, when the cam **322** moves to the outer side of the closing end **323***c* of the abutment member **323** and abuts against it, the cam **322** drives the extension portion **323***a* of the abutment member **323** to be embedded in the receiving slot, and at the same time, the first spring **324***a* and the second spring **324***b* are compressed, while locking the dosing apparatus **100** to prevent the dosing apparatus **100** from wobbling within the drive housing **315**.
(147) When the dosing apparatus **100** is locked, the first spring **324***a* and the second spring **324***b* are compressed to the same degree so that the elastic restoring force of the first spring **324***a* and the second spring **324***b* resisting the compression does not actuate the cam **322** to rotate, and thus the dosing apparatus **100** can be stably locked.
(148) When the toggle member **321** is pivoted towards the direction of the opening end **323***b* of the abutment member **323**, the cam **322** rotates about the camshaft **322***a* and moves from the outer side of the closing end **323***c* of the abutment member **323** to the outer side of the opening end **323***b* of the abutment member **323**.
(149) Since the width of the opening end **323***b* of the abutment member **323** is thinner, when the cam **322** moves to the outer side of the opening end **323***b* of the abutment member **323** and abuts

against it, the first spring **324***a* and the second spring **324***b* extend under the action of the elastic restoring force to restore from compression, and at the same time drive the extension portion **323***a* of the abutment member **323** to leave the receiving slot to release the locking of the dosing apparatus **100**, thereby facilitating smooth withdrawal of the dosing apparatus **100** from within the drive housing **315**.

(150) It should be noted that, in order to facilitate the illustration of the structure of the locking structure **320**, in FIG. **16**, only the state of the cam **322** being abutting against the abutment member **323** at the closing end **323***c* of the abutment member **323** is illustrated, and the actual state of the locking structure **320** when the dosing apparatus **100** is locked is not fully illustrated.

(151) A fourth aspect of embodiments of the present disclosure is to provide a dosing equipment **400**.

(152) Referring to FIG. **19**, in some embodiments, the dosing equipment **400** includes a container **211**, a dosing apparatus **100**, and a connector structure **220**. Wherein, the container **211** is adapted to hold the reagent and includes a container opening; the dosing apparatus **100** is at least partially disposed within the container **211** and is connecting with an interior and an exterior of the container **211** to convey the reagent within the container **211** at least outside of the container **211**; and the connector structure **220** is adapted to connect the container opening to the dosing apparatus **100**.

(153) In some embodiments, the connector structure **220** includes a connector **221** penetrating along the axial direction of the container opening, and a support member **222** and a bearing member **223** that are sequentially set within the connector **221**. The connector **221** is removably socketed around an outer periphery of the container opening; and there is a gap between the connector **221** and the support member **222** to receive the sidewall **211***a* of the container opening. The bearing member **223** is provided outside the dosing apparatus **100**. A first sealing ring **224** and a second sealing ring **225** are provided between the support member **222** and the bearing member **223**, and between the support member **222** and the sidewall **211***a* of the container opening, respectively, to seal-connect the connector structure **220** to the dosing apparatus **100** and the container opening, respectively.

(154) In some embodiments, the dosing apparatus **100** further includes a channel housing **124**, a restriction block **124***a* provided on an outer side of the channel housing **124**; the upper end surface of the bearing member **223** is recessed downwardly with respect to the upper end surface of the support member **222** and is adapted to receive as well as accommodate the restriction block **124***a*, to define a position at which the dosing apparatus **100** is coupled to the container **211**.

(155) In a specific implementation, the dosing apparatus **100** in this dosing equipment **400** may include the dosing apparatus **100** provided in the first aspect of the embodiment of the present disclosure.

(156) In some embodiments, the dosing equipment **400** further includes a drive mechanism **310**. The drive mechanism **310** is coupled to the dosing apparatus **100** to drive the dosing apparatus **100** to deliver the reagent. For example, a first spiral rod **112** in the dosing apparatus **100** may be driven by the drive mechanism **310** to rotate in a first direction and drive a second spiral rod **114** to rotate in a second direction.

(157) In particular embodiments, the drive mechanism **310** may include the drive mechanism **310** provided in the third aspect of embodiments of the present disclosure.

(158) In some embodiments, the drive mechanism **310** further includes a drive housing **315**, and the drive housing **315** has an opening that mates with the connector **221** to receive and connect the connector **221**.

(159) In some embodiments, the dosing equipment **400** further includes a lifting and turning structure **410**. The lifting and turning structure **410** is coupled to the dosing apparatus **100** to control the dosing apparatus **100**'s height and angle, so that the dosing apparatus **100** is adapted to align the reagent bottle receiving the reagent when outputting the reagent.

(160) It will be appreciated that when weighing and sampling the reagent, it is necessary to place

the reagent bottle below the second channel second opening **113***b* to receive the reagent. By providing the lifting and turning structure **410** to control the height and angle of the dosing apparatus **100**, the second channel second opening **113***b* can be smoothly aligned with the mouth of the reagent bottle for receiving the reagent.
(161) In the embodiment of the present disclosure, the lifting and turning structure **410** may be realized by any technical means known in the art.
(162) For example, the lifting and turning structure **410** may be pneumatically or hydraulically controlled to control the lifting and lowering of the dosing mechanism **100** or the drive housing **315**, as well as the height of the lifting and lowering.
(163) For another example, the lifting and turning structure **410** may include a bracket, and the angle of the dosing apparatus **100** may be controlled by the mounting angle between the dosing apparatus **100** or the drive housing **315** and the bracket.
(164) In some embodiments, since the container **211**, the dosing apparatus **100** may be connected to the drive housing **315** via the connector structure **220**, the lifting and turning structure **410** may also be connected to the drive housing **315** and be adapted to adjust the height and angle of the drive housing **315**, thereby controlling the height and angle of the dosing apparatus **100**.
(165) A fifth aspect of embodiments of the present disclosure is to provide a dosing system **500**.
(166) Referring to FIG. **20**, in some embodiments, the dosing system **500** may include a dosing apparatus **100**, a balance **510**, and a controller **520**. Wherein, the dosing apparatus **100** is adapted to deliver the reagent within the external container **211** into a reagent bottle outside the container **211**; the balance **510** is disposed below the reagent bottle and is adapted to weigh a mass of the reagent delivered into the reagent bottle; and the controller **520** is coupled to the balance **510** and the dosing apparatus **100**, respectively, and is adapted to regulate a speed of the movement of the conveying mechanism based on the mass weighed by the balance **510**.
(167) In particular embodiments, the dosing apparatus **100** in this dosing system **500** may include the dosing apparatus **100** provided in the first aspect of the embodiments of the present disclosure.
(168) In some embodiments, the conveying mechanism may include a conveyor belt **127**. In this case, the controller **520** may be coupled to a drive mechanism of the conveyor belt **127** and regulate a drive speed of the conveyor belt **127** via the drive mechanism.
(169) In other embodiments, the conveying mechanism may include a first spiral rod **112**. In this case, the controller **520** may be coupled to a drive mechanism **310** of the first spiral rod **112** and regulate the speed of the first spiral rod **112** through the drive mechanism **310**.
(170) In some embodiments, when the mass of the reagent delivered to the reagent bottle is not yet close to a target mass of the reagent, the first spiral rod **112** may be made to rotate at a high rotational speed so that the reagent is quickly delivered to the reagent bottle. And when the mass of the reagent delivered to the reagent bottle is close to the target mass of the reagent, the first spiral rod **112** can be made to rotate at a low rotational speed so that the reagent is slowly delivered to the reagent bottle, in order to prevent the problem of rapid delivery of the reagent, which leads to the problem of delivering an overdose of the reagent, from arising.
(171) It will be appreciated that the target mass of the reagent represents the mass of the reagent desired to be weighed.
(172) Adopting the above technical solution, by adjusting the rotational speed of the first spiral rod **112** in the dosing apparatus **100** in conjunction with the weighing feedback, the mass flow rate of the output reagent can be controlled, so as to better realize accurate weighing and sampling of the reagent.
(173) It has been proved by a large number of tests that by adopting the dosing apparatus **100**, the dosing set **200**, the dosing head **300**, the dosing equipment **400**, and the dosing system **500** provided in the embodiments of the present disclosure, the mass flow rate of the output reagent is controllable and remains constant at a constant rate of sampling, and the accuracy of the sampling can thus be effectively ensured.

(174) Although specific embodiments of the present disclosure have been described above, these embodiments are not intended to limit the scope of the disclosure of the present disclosure, even where a single embodiment is described only in relation to a particular feature. The examples of features provided in the present disclosure are intended to be illustrative and not limiting, unless a different representation is made. In specific embodiments, the technical features of one or more dependent claims may be combined with the technical features of the independent claims according to practical needs and where technically feasible, and the technical features of the corresponding claims may be combined in any appropriate manner rather than only by the particular combinations enumerated in the claims.
(175) Although the invention is disclosed as above, the invention is not limited thereto. Any person skilled in the art may make various changes and modifications without departing from the spirit and scope of the present disclosure, and therefore the scope of protection of the present disclosure shall be subject to the scope limited by the claims.

Claims

1. A dosing equipment (**400**), comprising: a container (**211**) which is adapted to hold reagent and includes a container opening; a dosing apparatus (**100**), which is at least partially provided within the container (**211**) and is connecting with an interior and an exterior of the container (**211**) to convey the reagent from the interior of the container (**211**) to the exterior of the container (**211**); and a connector structure (**220**) which is adapted to connect the container opening and the dosing apparatus (**100**); wherein the connector structure (**220**) includes a connector (**221**); and the connector structure (**220**) penetrates the container along an axial direction of the container opening, a support member (**222**) and a bearing member (**223**) sequentially socketed within the connector (**221**); wherein the connector (**221**) is socketed on an outer periphery of the container opening; wherein a gap is provided between the connector (**221**) and the support member (**222**) to receive a sidewall (**211***a*) of the container opening; wherein the bearing member (**223**) is provided outside the dosing apparatus (**100**); wherein a first sealing ring (**224**) and a second sealing ring (**225**) are provided between the support member (**222**) and the bearing member (**223**) and between the support member (**222**) and the sidewall (**211***a*) of the container opening, respectively, so as to seal-connect the connector structure (**220**), the dosing apparatus (**100**), and the container opening, respectively; wherein the dosing apparatus (**100**) comprises: a first channel (**111**), wherein the first channel (**111**) has a first channel first opening (**111***a*) connecting with the container (**211**) to allow the reagent when held inside the container (**211**) to enter the first channel (**111**), and a first channel second opening (**111***b*) which is in connection with outside of the first channel (**111**) to allow the reagent to leave the first channel (**111**); and at least one first spiral rod (**112**) provided within the first channel (**111**), wherein the least one first spiral rod (**112**) is configured to be driven to rotate in a first direction, so as to deliver the reagent from the first channel (**111**) to the first channel second opening (**111***b*), and wherein a mass flow rate of the reagent leaving through the first channel second opening (**111***b*) is controlled by adjusting a rotational speed of the at least one first spiral rod (**112**).
2. The dosing equipment (**400**) according to claim 1, wherein the dosing apparatus (**100**) further comprises a channel housing (**124**), and a restriction block (**124***a*) disposed on an outer side of the channel housing (**124**); wherein an upper end surface of the bearing member (**223**) is recessed downwardly with respect to an upper end surface of the support member (**222**), so as to be suitable for receiving and accommodating the restriction block (**124***a*), thereby defining a position at which the dosing apparatus (**100**) is coupled to the container (**211**).
3. The dosing equipment (**400**) according to claim 1, wherein the dosing apparatus (**100**) further comprises a second channel (**113**) and at least one second spiral rod (**114**) provided within the second channel (**113**); wherein the second channel (**113**) has a second channel first opening (**113***a*)

in connection with the first channel second opening (**111***b*) to allow the reagent within the first channel (**111**) to enter the second channel (**113**), a second channel second opening (**113***b*) in connection with the outside to allow the reagent to leave the second channel (**113**), and a second channel third opening (**113***c*) in connection with the container (**211**) to allow the reagent to leave the second channel (**113**) and return to the container (**211**); wherein the at least one second spiral rod (**114**) is in meshing transmission connection with the at least one first spiral rod (**112**) to rotate in a second direction as driven by the first spiral rod (**112**), to convey the reagent within the second channel (**113**) to the second channel second opening (**113***b*) and the second channel third opening (**113***c*); and wherein the second direction is opposite to the first direction.

4. The dosing equipment (**400**) according to claim 3, wherein the first channel first opening (**111***a*) is provided in a mid-side portion of the first channel (**111**); the first channel second opening (**111***b*) is provided in a lower side portion of the first channel (**111**); the second channel first opening (**113***a*) is provided in a lower side portion of the second channel (**113**); the second channel second opening (**113***b*) is provided at a bottom end of the second channel (**113**); and the second channel third opening (**113***c*) is provided at an upper side portion of the second channel (**113**).

5. The dosing equipment (**400**) according to claim 3, wherein a tip of the at least one first spiral rod (**112**) extends outside the first channel (**111**); wherein a tip of the at least one second spiral rod (**114**) extends outside the second channel (**113**); wherein the dosing apparatus (**100**) comprises an end-face cam (**115**) coaxially connected with the tip of the at least one first spiral rod (**112**) and a first gear (**116**) coaxially connected with the tip of the at least one second spiral rod (**114**); wherein the end-face cam (**115**) is in meshing transmission connection with the first gear (**116**); and wherein the at least one first spiral rod (**112**) is adapted to be driven by a driving apparatus (**310**) to rotate in the first direction, thereby driving the at least one second spiral rod (**114**) to rotate in the second direction.

6. The dosing equipment (**400**) according to claim 3, further comprising a driving apparatus (**310**); wherein the driving apparatus (**310**) is connected to the first spiral rod (**112**) and is adapted to drive the at least one first spiral rod (**112**) to rotate in the first direction, and to drive the at least one second spiral rod (**114**) through the at least one first spiral rod (**112**) to rotate in the second direction at the same time.

7. The dosing equipment (**400**) according to claim 6, wherein the driving apparatus (**310**) further includes a drive housing (**315**); wherein the drive housing (**315**) has an opening matching the connector (**221**) so as to accept and connect to the connector (**221**).

8. The dosing equipment (**400**) according to claim 7, further comprising a lifting and turning structure (**410**); wherein the lifting and turning structure (**410**) is coupled to the dosing apparatus (**100**) to control a height and an angle of the dosing apparatus (**100**); and wherein the dosing apparatus (**100**) is adapted to be aligned with a reagent bottle for receiving the reagent when the dosing apparatus (**100**) is outputting the reagent.

9. The dosing equipment (**400**) according to claim 8, wherein the lifting and turning structure (**410**) is connected to the driving apparatus (**100**) through the drive housing (**315**).